

Urban agriculture growth in US cities

In the United States, urban agriculture is growing as a result of increased availability of unused land and innovative development; the growth of farms and community gardens improves the ability of community members to cope with social and environmental change. But what will make urban agriculture sustainable?

When the mechanics that occupied the quarter-acre lot in the middle of northwest Washington DC closed their business and abandoned the property, the neighbours had much to fear. The city did not know what else it could do with a small plot of land filled with disassembled automobiles; residents worried about the rusting pile of junk and potential for crime. But the non-profit organization Washington People and Parks saw it as an opportunity for a community garden. The city later cleared tax liens and sold the plot for US\$1 to North Columbia Heights Green, a neighbourhood association, and an urban farm was born.

Ten years later, Kate Tully, an agricultural ecologist at the University of Maryland and a resident of the neighbourhood, farms North Columbia Heights Green with 25 area residents. Like many urban farms, Tully's group grows so much food they cannot consume it all and they share surplus food with a soup kitchen, Martha's Table¹. The urban garden contains ten large raised beds that form a semi-circle and residents grow the produce collectively. Planted together using methods of agroecology, crops enhance each other's growth, protect each other from pests, and with cover crops they add nutrients to the soil. The result has led to one of the most productive urban farms in the city, as well as attracting 100 participants each year through planting, maintenance, and pick-your-own initiatives. The growers do not buy vegetables during the summer season because they produce around 16 kg of fresh vegetables each week. Research suggests that urban agriculture increases fruit and vegetable consumption among those who are active growers².

Urban agriculture has always been a part of city life in the United States, whether in the development of frontier towns or as a strategy to improve urban conditions. Depression-era relief gardens, victory gardens grown during the First World War to augment food supply, and post-industrial community gardens are examples³. Urban agriculture includes a range of cultivated areas, from urban farms to community gardens, and has been cropping up in vacant land, rooftops, back yards, warehouses, and shipping containers from Baltimore to



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Oakland⁴. Growing food in cities has been on the rise since the 1980s, when an increase in access to urban land made it easier to establish small farms and gardens and the social, political, and economic climate accelerated interest.

In low-income areas of cities across the United States, supermarkets have withdrawn to more affluent areas, leaving residents with few fresh produce offerings — often expensive and of poor quality — at small corner stores. Well-positioned urban agriculture can complement food access strategies in these urban food deserts, where it is extremely difficult to buy healthy food^{3,5}. A hamburger at a fast-food restaurant can cost US\$1, and yet a small package of carrots or sliced apples costs three times as much. Growing produce goes a long way toward coping with food deficiencies, says Tully.

Revamping unused land to build social cohesion, improve health and wellbeing, and add to agro-biodiversity is an appealing proposition, but the farms often face multiple barriers such as extremes in soil alkalinity or acidity, soil contamination, and access to water. North Columbia Heights Green worked around these problems by

growing crops in raised beds with soil purchased from garden stores and running a hose to the garden from Tully's apartment.

Since 2008, state and city planners and urban-development agencies have been actively promoting urban agriculture after an increase in the availability of unused land, and innovative development emerged as a way to revitalize brownfield sites and increase community ties after economic downturn. Detroit has become the poster child for sustainable urban redevelopment, with the Michigan Urban Farming Initiative (<http://www.miufi.org/>) transforming two square blocks of Detroit's North End into what it calls "America's first sustainable urban agrihood".

Once set in motion, urban agriculture requires coordinated local efforts to gain momentum and become sustainable. Cities keep interest going by conducting land inventories, establishing food policy councils, and revising policies to accommodate the new land-use changes, for example⁶.

Small community gardens, urban farms that span several city blocks, and intensive indoor hydroponic and aquaculture facilities

are all examples of urban agriculture. Urban agriculture continues to be a planning priority for cities, which now seek to measure and understand: the social and environmental benefits they provide; whether fresh food is getting to where it is needed; and the economic development potential of cultivating disinvested spaces.

Some of the residents of Washington DC's North Columbia Heights most in need are El Salvadoran and Eritrean immigrants, says Tully. While adults from these communities have not been involved in that urban farm, youth from these families regularly help with the farm's chores and bring bags of food back to their homes. In this way, urban agriculture improves cross-cultural and cross-generational linkages, as seen elsewhere^{7,8}.

Urban agriculture embodies sustainability by promoting ecological and social benefits, and, potentially, economic ones. However, most analyses do not show urban farms solving food security or providing economic opportunities at the city scale. Urban agriculture can increase urban biodiversity and green space by acting as islands in a sea of concrete, attracting native plants, pollinators, and a variety of small animals. For instance, gardens set in urban areas have higher ladybird abundance and richness⁹. They also facilitate drainage of water and reduce the urban heat island effect¹⁰. Urban agriculture can also increase resilience to food shocks by reducing produce costs during the growing season and overcoming any disruptions in distribution. Urban farms and community gardens beautify communities, enhance relationships, and serve as a bridge to access social services, including nutritional education, youth development, and job training^{11,12}.

For Monika Egerer of the University of California, Santa Cruz, and Elsa Anderson of the University of Illinois at Chicago, urban farms have inspired a new research pursuit to understand how gardens may act as nodes for both environmental and social connectivity — how they can be food islands for pollinators and meeting places for community groups, for instance. Egerer and Anderson are working with a team of researchers to catalogue the biophysical characteristics of the urban farms in Baltimore, Chicago and New York to develop a joint understanding of environmental aspects, including soil contamination, the ability to capture water, pollinator activity, and the ability to reduce the heat island effect; and also social characteristics of community support such as whether non-profits, churches, community centres, schools or social amenities exist in an area near the urban farms. Where they see holes

in the landscape, such as a low number of community services or green spaces, Egerer and her team seek to advise planners to consider starting an urban farm or community garden to serve those social and ecological functions. In this way, Egerer says she hopes her research team can provide a spatial analysis of urban agriculture that will not only “provide a bird's eye view of where urban farms are located, but also a worm's eye view of who is actually using the urban land for growing food”.

When urban agriculture is officially authorized by a city, through sanctions or permits, it can be counted more easily, providing a key to measuring it. Sanctioned means it has received some kind of government or non-governmental organization support to license and establish itself. These sanctioned spaces range from urban farms for business-oriented, commercial food production to community gardens, such as Tully's Washington DC farm, and other gardens on plots that are rented or allotted to individual growers. Unsanctioned spaces mean there is no sponsorship or support by an organization, and these are typically individuals growing in backyards on a patch of green for their home consumption.

Urban agriculture promoters claim that it is emerging as an agent of change and an important tool for sustainability, aided in part by its deep social and ecological value. In Milwaukee, where 53% of black men were unemployed in 2009, the non-profit urban farm Growing Power created 150 jobs for low-income residents, including prisoners, to improve both the social and economic wellbeing of inner-city, low-income residents¹³.

In New York City, promoters are a strategic part of the city's parks department and have a background in urban design. The city is home to the largest number of urban gardens in the United States. Green Thumb, an organization that has led efforts to develop urban agriculture in New York City since 1978, oversees more than 550 sanctioned urban gardens in all five boroughs. The group started when activists and volunteers cleaned up vacant lots so they could plant community gardens. Now, community gardens in New York cover 100 acres of land, primarily in low-income areas, and almost three quarters of the gardens include an urban agriculture component that grows food.

With strong support and knowledge that the gardens have a long start-up time to get off the ground to be socially sustainable, Green Thumb director Bill LoSasso, an urban planner, is now developing a rubric for positioning new urban gardens in low-income neighbourhoods that need them most. He seeks to focus on places that do not

already have a community garden, lack open space, have high obesity rates, are reporting low consumption of fruits and vegetables, and where people report they lack food for regular meals, based on data from the health department, food banks, city agencies, and non-profits.

LoSasso says that the urban gardens become *de facto* outdoor community centres that promote sustainable cohesion and community ties¹⁴. “Gardens cut across age and race and financial barriers by bringing people together to share stewardship of open space”, he says. “They are a unique space in bringing people together even with the tensions that can come with gentrification. Gardens are neutral space where everyone participate, and they are having a positive impact in keeping neighbourhoods strong.”

While business-oriented urban farms tend to be found in larger cities, it is unclear how the size of a city determines the number of urban farms and gardens in it; factors besides city size are important, too. “A lot has to do with the city itself, and the space available”, says Anderson. Baltimore's history of development with entire blocks built with row houses has meant that, with disinvestment in the city, entire city blocks have been torn down, resulting in abandoned lots that span whole blocks. This has prompted larger urban gardens, on average, although individual small plots do occur. In Chicago, there are 150 sanctioned urban farms that are smaller and more uniform in size, says Anderson.

When it comes to economic opportunities, about 5–10% of produce grown in urban spaces is for-profit and sold to markets, either direct to consumer or through purchase agreements with institutions. Urban Pastoral in Baltimore, for example, is led by a team of agriculture developers. Their business model of urban farming is through controlled environment agriculture in buildings, shipping containers, or unused spaces. Using principles of integrated vertical growing and design thinking to increase environmental efficiencies, their focus remains on urban development and all their production is through long-term purchase agreements with local restaurants, schools, and institutions, with the potential for adding large, commercial-scale, controlled-environment urban facilities. Similarly, AeroFarms in Newark, New Jersey, operates out of a former steel mill, and has developed a proprietary growing system where plants grow vertically in stacks from fabric under artificial light and use a fraction of the water of conventional gardens¹⁵.

Finally, urban agriculture can play a unique role in sustainability by linking

growers and consumers in a common system that is mutually supportive. Suburban and peri-urban farmers gain urban market share and improved viability through seasonal subscription-oriented community-supported agriculture, purchase agreements with restaurants, and farmers' markets. Consumers that buy from local producers benefit from fresh produce. The environment benefits from biodiverse ecosystems near cities, and farmers close the sustainability loop in this system by benefiting from the rewards of selling in cities.

The post-industrial trend of urban agriculture in cities can bring important shifts to social and ecological systems in the United States. As the growth of farms and gardens caused by the increased

availability of unused land brings change, formerly vacant lots can shift from eyesores to attractive and productive green spaces, providing an array of benefits that have the potential to grow the sustainable cities of the future. 

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